

Anant Pareek

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Machine Learning Engineer specializing in Software as a Medical Device (SaMD) systems, with research interests in computational psychology and affective AI. Developing clinically validated diagnostic models while investigating how intelligent systems can better understand and adapt to human behavior.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer (Intern: Oct 2025 – Feb 2026)

Oct 2025 – Present

Lenek Technologies Private Limited

Pune, India (Remote/Hybrid)

- Secured early promotion to full-time engineer within four months by successfully prototyping foundational computer vision pipelines for radiological diagnostics.
- Developing and optimizing Software as a Medical Device (SaMD) AI models for high-burden pulmonary and cardiac conditions, specializing in Chest Radiographs (CXRs) and ECG analysis.
- Architecting deep learning pipelines to process and classify complex medical imaging, ensuring high diagnostic accuracy and system reliability in clinical settings.
- Co-authored and facilitated a large-scale, multi-source clinical validation study to rigorously evaluate the platform's efficacy, currently under peer review at the International Journal of Research in Medical Sciences.

RESEARCH & PROJECTS

Stress-Testing SLM Personality Architectures | *Independent Research, LoRA, Unsloth*

Mar 2026 – Present

- Investigating the psychological collapse of Small Language Models (SLMs) under adversarial conversational pressure, specifically mapping the degradation of Big Five traits.
- Engineered a multi-agent adversarial pipeline to generate thousands of stress-test dialogues, utilizing Beta-distributed trait sampling to eliminate LLM "middle bias" and enforce psychometric extremes.
- Identified critical RLHF alignment constraints (e.g., Extraversion flatlining) and currently scaling a dataset of 5,000 high-quality "survivor" dialogues for parameter-efficient fine-tuning via Unsloth.

Multi-Model Adaptive Personality AI | *Custom Model Quantization, API Deployment*

[\[GitHub\]](#)

- Engineered a custom RobertaForRegression model to resolve critical training instability in transformer-based regression, boosting the EmoBank R^2 score from a baseline of 0.13 to a state-of-the-art 0.47.
- Designed and deployed a microservices architecture, hosting five specialized models (including a Gemma-2B "Personality Brain") as independent Gradio APIs on Hugging Face Spaces.
- Built a central Streamlit dashboard to orchestrate the backend APIs, providing an interactive interface for live, multi-model psychological analysis.

EEG Motor Imagery Classification | *Deep Learning, MLOps, TensorFlow*

[\[GitHub\]](#)

- Engineered an end-to-end deep learning pipeline to classify motor imagery tasks from raw EEG data, achieving 74.8% accuracy on the held-out test set.
- Implemented a Dockerized environment and CI/CD via GitHub Actions for reproducible, scalable experimentation.

PUBLICATIONS & PRE-PRINTS

A Retrospective Multi-Source Clinical Validation of Lenek Intelligent Radiology Assistant: An Artificial Intelligence-Based Chest Radiograph Screening and Triage System for High-Burden Pulmonary and Cardiac Conditions in India (2026)

Submitted to the International Journal of Research in Medical Sciences (under peer review).

Pre-print available at: [\[medRxiv\]](#)

A Multi-Component AI Framework for Computational Psychology: From Robust Predictive Modeling to Deployed Generative Dialogue (2025)

Submitted to the Expert Systems With Applications Journal (under review).

Pre-print available at: [\[arXiv\]](#)

EDUCATION

St. Joseph's University <i>Bachelor of Arts, Psychology and Optional English</i>	<i>July 2022 – May 2025</i>
TiHAN, IIT Hyderabad <i>Specialized AI/ML Program with Drone Tech</i>	<i>July 2025 – May 2026</i>
WorldQuant University <i>Applied AI Lab, Deep Learning for Computer Vision</i>	<i>July 2025 – July 2026</i>

TECHNICAL SKILLS

Languages: Python, R, SQL
Frameworks: PyTorch, TensorFlow, Keras, FastAPI, Flask, React, Unsloth
AI/ML Tools: Hugging Face (Transformers, PEFT, Spaces), LoRA, Model Quantization, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Gradio
MLOps/Dev Tools: Git, Docker, GitHub Actions, CI/CD, Google Cloud Platform, AWS, VS Code

CERTIFICATIONS

IBM AI Developer Professional
IBM Deep Learning with PyTorch, Keras, TensorFlow
IBM Data Science Professional
Google Data Analytics Professional
UC Davis Market Research Specialization
APA Psychological Research Specialization